

Sheridan College

Institute of Technology and Advanced Learning

INFO49402 - Capstone Project Phase 2

**Reply All: A Cybersecurity Awareness Gamification
Platform**

Final Project Report

Group Members

Troy Patrick

Vladimir Gappasov

Ayinkaran Ravindran

Capstone Advisor: Syed Tanbeer

Submitted: Week 14, 2026

Table of Contents

Table of Contents	2
Glossary of Terms and Acronyms	4
Abstract	5
Problem Statement	6
The Issue of Concern	6
Importance to Information System Security	6
Project Goals and Objectives	6
Project Contribution	7
Prior Work and Relevant Literature	8
Gamification and Learning	8
Human Factor in Cybersecurity	8
Existing Platforms and Their Limitations	8
Game Engine Selection	9
Chosen Solution Details	10
Overview and Justification	10
Scope and Team Contributions	10
Issue System	10
NPC System and Patience Mechanics	11
Critical Breach Lockdown Event	12
HUD UI: How to Keep the Player Informed	12
Popup UI: iPad-Style Tablet	13
Audio System	13
Cutscene System	14
Reflection Report	14
What Worked	15
What Did Not Work	15
Results and Outcomes	17
Playtesting Methodology	17
Engagement and Difficulty	17
Learning Outcomes	17
Lockdown Event Outcomes	17
Session Outcomes	17
Reflection Report Engagement	17
Observations	18
Conclusions and Future Work	19
Conclusions	19
Future Work	19
Bibliography	21
Appendices	22
Appendix A: Project Links	22
Appendix B: Full Issue Catalog	22
Appendix C: Playtesting Data Summary	22
Tester Demographics	22

Engagement Survey Results	22
Difficulty Ratings (1 = easiest, 5 = hardest)	22
Session Outcomes	22
Lockdown Event	23
Reflection Report	23
Appendix D: Individual Contribution Summary	23

Glossary of Terms and Acronyms

API - Application Programming Interface: a set of protocols that allows software components to communicate.

BEC - Business Email Compromise: a cyberattack where an attacker impersonates an executive to authorize fraudulent transactions.

C2 - Command and Control: infrastructure used by attackers to remotely manage compromised systems.

GDScript - Godot Script: a Python-like scripting language native to the Godot game engine.

IT - Information Technology: the use of computing infrastructure to create, process, store, and exchange data.

MFA - Multi-Factor Authentication: a security mechanism requiring more than one verification factor to grant access.

NPC - Non-Player Character: a game character controlled by the game engine rather than a human player.

QTE - Quick Time Event: a gameplay mechanic requiring fast player responses to on-screen prompts within a time limit.

RDP - Remote Desktop Protocol: a Microsoft protocol enabling remote graphical access to another computer.

SFX - Sound Effects: audio feedback cues triggered by in-game events.

SIEM - Security Information and Event Management: software that aggregates and analyzes security data across an organization.

SOC - Security Operations Center: a centralized function that monitors, detects, and responds to security incidents.

UI - User Interface: the visual layer through which a user interacts with software.

VPN - Virtual Private Network: technology that creates an encrypted tunnel between a user device and a network.

Abstract

Cybersecurity threats that are driven by human error continue to grow in frequency and cost despite significant organizational investment in beefing up firewalls and various complex protections. The persistent effectiveness of social engineering attacks, such as phishing, social engineering, and credential attacks against employees who have completed conventional awareness training points to a fundamental inadequacy in passive instructional methods. This paper presents Reply All, a 2D top-down pixel art game developed in Godot 4.7 that addresses this gap by placing learners in an active, consequence-driven simulation of workplace cybersecurity incident response.

Players control an IT technician navigating a pixel art office environment where non-player characters raise security incidents in real time. Incidents span 36 unique scenarios across four threat levels, from low-severity issues such as forgotten passwords and unlocked workstations to critical threats including active network intrusions, ransomware deployments, command-and-control beacon detections, and live data exfiltration. The platform implements a dynamic danger bar that accumulates as incidents go unresolved, a patience system that escalates unattended tickets automatically, a critical breach lockdown mechanic requiring the player to physically clear every NPC within 45 seconds under time pressure and intense adaptive music that shifts depending on the situation, and a post-shift endgame Reflection Report that displays every wrong answer alongside the correct response and a plain-language explanation to all users to further reflect on choices and mistakes made.

Playtesting across 9 participants reflected multiple interesting data points. 6 of 9 testers engaged with the Reflection Report without prompting, and 8 of 9 reported learning something that they did not previously know. The project demonstrates that consequence-driven gameplay is a viable, engaging, and better teaching mechanism for delivering cybersecurity awareness content to the public and to general office workers who do not possess in-depth knowledge of cybersecurity or cyber-attack techniques or procedures.

Problem Statement

The Issue of Concern

Cybersecurity incidents that are caused by human behavior remain one of the most significant and least-solved challenges in information system security. According to SentinelOne, the human element contributed to 70 percent of all breaches analyzed, with phishing, pretexting, and credential theft representing the dominant initial access vectors [1]. Despite widespread adoption of mandatory security awareness training programs, employees continue to fall victim to the same categories of attack year over year.

The current problem is that a current everyday working adult's technical knowledge has been far eclipsed by the speed of technological growth. People and employees alike often use technology with a lack of understanding of the proper safety training needed to navigate them safely. It is easy in a controlled environment to tell someone not to click on a link because it is dangerous, however it has been shown that possessing theoretical knowledge does not necessarily translate into applying it effectively in real-world decision-making. Employees who complete a training module and score 90 percent on a knowledge check still click malicious links within weeks, because passive instructional formats do not build the rapid recognition and reflexive response that real incidents demand. Research by Roer and Petric found that employees who completed standard awareness training showed minimal behavioral change in phishing simulations conducted weeks after the training session [2].

Importance to Information System Security

The human layer is the most consistently exploited attack surface in enterprise security. Unlike software vulnerabilities, which can be patched, human susceptibility to social engineering and authority cues is persistent and not addressable through technical controls alone. SentinelOne placed the average cost of a breach at USD 4.88 million, with phishing-initiated breaches among the most expensive due to the depth of access attackers achieve before detection [1].

Organizations that address only the technical attack surface while relying on passive training for the human layer are leaving a structural gap that attackers systematically exploit.

Gamification of training and learning has been proposed as a mechanism for improving engagement and behavioral change in security training contexts. Active learning formats, where participants make decisions under simulated consequence, consistently outperform passive formats in knowledge retention and behavioral compliance studies [3]. However, existing gamified security platforms either target technical security professionals or focus on simple quiz mechanics without simulating the workplace context in which real decisions occur.

Project Goals and Objectives

The goals of this project were to:

- Design and implement a playable cybersecurity awareness game that covers a broad and realistic range of incident types encountered in a typical office environment.
- Create 36 unique cybersecurity scenarios categorized by threat level, each grounded in real attack techniques with technically accurate response options and post-answer explanations.
- Implement a consequence-driven game loop in which player decisions directly affect game state.
- Build a mechanic that simulates active incident response under genuine time pressure to create memorable, high-stakes learning moments.
- Encourage players to instinctively prioritize multiple threats using a mental triage process, focusing on mitigating the highest-risk incidents first.
- Allow players to reflect on mistakes and have opportunities to learn from experience through organic gameplay and visual post-action reports.

Project Contribution

Reply All contributes a playable, open-source cybersecurity awareness prototype that demonstrates the viability and practicality of active, consequence-driven gameplay for general organizational security education. The project fills a specific gap in the existing landscape: no freely available game targets non-technical employees in a realistic workplace setting with technically accurate incident content and genuine gameplay consequences. The platform also demonstrates a complete multi-branch collaborative development workflow in a game engine context, with clearly documented individual contributions to content, systems, audio, and visual design.

Prior Work and Relevant Literature

Gamification and Learning

Deterding et al. established the foundational definition of gamification as the application of game-design elements in non-game contexts, identifying points, narrative consequence, feedback loops, and progression as the primary drivers of engagement and intrinsic motivation [4]. In educational settings, gamification has been shown to improve both time-on-task and knowledge retention, particularly when the game mechanics are aligned with the learning objectives rather than added as superficial reward layers on top of unmodified instruction.

Mayer conducted a comprehensive meta-analysis of educational game research and found that game-based learning produced significantly higher retention outcomes than lecture-based instruction across science, mathematics, and procedural domains [5]. The effect was most pronounced for procedural knowledge, which is the category most relevant to cybersecurity incident response. Knowing that a phishing email should not be clicked is declarative knowledge; recognizing and acting on a phishing email under social pressure is procedural, and it is the procedural gap that Reply All specifically targets.

Human Factor in Cybersecurity

Heartfield and Loukas examined the cognitive mechanisms underlying social engineering susceptibility and identified three primary factors that cause employees to bypass skepticism: time pressure, perceived authority, and contextual plausibility [6]. Their work is directly reflected in the design of Reply All's critical threat scenarios, particularly the Business Email Compromise and MFA Push Bombing incidents, which exploit all three factors simultaneously. The critical breach lockdown mechanic introduces genuine time pressure to the gameplay, mirroring the conditions under which real incident response decisions are made.

Workman studied behavioral compliance after experiential versus informational security training and found that experiential learning, defined as training where participants actively respond to simulated attacks and receive consequence feedback, produced significantly higher post-training behavioral compliance than information-only instruction [7]. This finding provides the primary theoretical justification for the gameplay loop in Reply All, where every wrong answer produces an immediate in-game consequence rather than a static score deduction.

Existing Platforms and Their Limitations

Several existing platforms occupy the learning and gaming-adjacent space to Reply All. KnowBe4 and Proofpoint offer phishing simulation tools that measure susceptibility but do not teach response; they function as assessment instruments rather than training mechanisms [8]. Immersive Labs and Hack The Box provide gamified technical security training, but both

platforms assume a baseline of security professional knowledge that makes them inaccessible to general organizational employees. CyberCIEGE, developed by the Naval Postgraduate School, is the closest academic precedent, offering a resource management game incorporating security policy decisions, but its interface complexity and academic framing limit adoption in organizational training contexts [9]. There are also multiple games such as Hacknet and Uplink which gamify the ability for someone to gain technical skills rather than practical problem-solving knowledge.

Reply All is specifically designed to fill the gap between these categories: a game accessible enough for any employee to begin playing immediately, grounded in realistic workplace scenarios rather than abstract puzzles, with genuine consequence mechanics that create the kind of pressure and context that research indicates is necessary for behavioral transfer. This was made with the idea in mind that instead of teaching technical knowledge and theory it was more important to bring awareness to the actual everyday problems that one might encounter.

Game Engine Selection

We chose Godot 4.6/4.7 as the development environment. Godot is an open-source game engine with an active community and a Python-like scripting language, GDScript, that provides low friction for developers with programming backgrounds but limited prior game development experience. Moreno-Ger et al. identified ease of authoring, asset flexibility, and scripting expressiveness as the primary criteria for educational game development platform selection [10]. Godot 4 satisfies all three and adds zero commercial licensing cost, making it appropriate for an open academic project intended to be freely distributed.

Chosen Solution Details

Overview and Justification

Reply All is a 2D top-down office environment game built in Godot 4.6/4.7 using GDScript. The player character is an IT technician who moves through a pixel art office, responds to cybersecurity incidents raised by non-player characters, and must manage a danger bar that increases as incidents go unresolved or are answered incorrectly. The shift ends either when the three-minute timer expires or when the danger bar reaches 100 percent, representing a company-wide security compromise. The format was chosen because it provides the active, consequence-driven decision-making context that research indicates is necessary for procedural learning, while keeping the interface accessible to players with no prior gaming experience.

A core gameplay design philosophy was to develop the player's ability to assess multiple concurrent cybersecurity threats and instinctively prioritize them based on their potential impact. Rather than explicitly instructing the player about which incident is most critical, the game encourages them to form an internal triage process, continuously evaluating and re-evaluating competing threats as new incidents emerge. Through repeated exposure to these decision-making scenarios, players learn to allocate their limited time and attention to the threats that pose the greatest risk, while understanding that not every incident requires an immediate response. We believe this reinforces the cybersecurity principle of threat mitigation by teaching players to prioritize high-impact incidents and make effective decisions under pressure.

Scope and Team Contributions

The project was divided across multiple contribution branches in GitHub, each owned by one team member and created based on purpose:

- Troy Patrick: Sound effects and adaptive music system, iPad-style tablet popup UI with slide-in animation, intro and game over cutscene system, Options menu with volume and shift length controls, critical breach lockdown event, 24 new cybersecurity issue .tres resource files, and the Reflection Report screen.
- Vladimir Gappasov: Scoring and points system, threat level danger bar with colour coding, shift timer display, scoreboard integration, and issue content contributions.
- Ayinkaran Ravindran: Character sprite improvements, map creation and Y-sorting for visual depth perception, NPC urgency timing per role type, escalation consequence system, walking animation refinements, HUD design, and notification system.

All branches were merged into the main branch via pull requests tracked in GitHub, providing a complete audit trail of individual contributions as well as the order of features implemented and refined.

Issue System

The core content of Reply All consists of 36 cybersecurity incident scenarios stored as Godot resource files (.tres) using a shared CyberIssue class. Each resource contains: the issue name, a description visible in the debug view, an employee message displayed in the tablet popup written from the perspective of the NPC with authentic workplace dialogue, a threat level integer from 1 to 5, an urgency float controlling the patience drain rate, a correct answer index, three answer options, an explanation shown after answering, and an optional escalation resource reference.

The 36 issues are categorized as follows:

- Low threat (1-2): Forgot Password, Weak Password, Public Wi-Fi Risk, Unlocked Workstation, Shared Account Password, Tailgating into Restricted Area.
- Mid threat (3): Phishing Email, Social Engineering Call, Sensitive Documents on Printer, Suspicious USB, Outdated Software Patch, Malicious Email Attachment, Exposed RDP Port, Public Cloud Storage Bucket.
- High threat (4): Data Leak, Unauthorised Access, Ransomware Alert, Zero-Day Vulnerability Disclosure, Insider Data Theft, Business Email Compromise.
- Critical threat (5): Active Network Intrusion, Credential Database Dump, Ransomware Actively Encrypting, Supply Chain Compromise, MFA Push Bombing, C2 Beacon Detected, Live Data Exfiltration.

Issues spawn using a weighted pool: 35 percent critical, and the remainder split between high and normal pools. Critical issues are separated into their own pool because a wrong answer on a threat level 5 issue triggers the lockdown event rather than a standard danger penalty.

NPC System and Patience Mechanics

Eleven NPCs are placed across the office map, each assigned a role type with individual patience and drain rate configuration. The `ROLE_CONFIG` dictionary in `employee.gd` defines these parameters per role. IT NPCs have patience of 85 and a drain multiplier of 1.2, reflecting moderate urgency. CEO characters have patience of 120 with a drain rate of 1.0 after tuning. Interns have patience of 130 with a drain rate of 0.7, representing their higher tolerance. Receptionists and general workers sit at patience 100 with drain 1.0.

Issue urgency values, stored in each .tres file, are capped at 0.35 following playtesting feedback that higher values caused some tickets to expire in under 10 seconds, insufficient for a player to read the scenario. The effective time before escalation is calculated as $\text{patience} \div (\text{urgency} \times 10 \times \text{drain multiplier})$. At the tuned values, the minimum readable time for a CEO ticket is approximately 34 seconds, while intern tickets provide over a minute.

When patience reaches zero, `escalate()` fires on the employee. This clears the issue, adds danger proportional to threat level times 8, deducts score, and records the issue in the Reflection Report as a timed-out wrong answer, ensuring that escalated jobs appear in the post-session review even though the player never actively answered them incorrectly.

Critical Breach Lockdown Event

The lockdown event is the highest-stakes mechanic in the game and the primary differentiator from existing cybersecurity awareness tools. When a player selects the wrong answer on a threat level 5 issue, `issue_popup.gd` calls `GameManager.trigger_lockdown()` directly rather than routing through the employee node. This was a deliberate design decision: routing through the employee introduced a race condition where the employee's patience could expire and clear `current_issue` between the wrong answer being selected and `trigger_lockdown()` being called, silently cancelling the event.

When the lockdown fires:

- A red breach alert panel appears at the top of the screen showing a 45-second countdown and a progress bar tracking cleared stations.
- The background music switches to an intense 160 BPM minor-key track with dissonant tritone stabs generated procedurally using Python's wave module.
- `show_lockdown_alert()` fires on every NPC, enlarging their exclamation mark to 1.8x scale and setting it to bright red.
- The player must physically walk to each of the 11 NPCs and press E to clear their station. A per-session cleared set prevents spam-clearing by tracking which specific employee instances have been visited.
- Clearing all NPCs in time awards 500 points, restores the normal music, and hides all lockdown alerts. Failing the 45-second countdown sets danger to 100 and ends the shift immediately.

HUD UI: How to Keep the Player Informed

The HUD keeps track of the player score and time in the top corners of the screen, and the threat bar at the bottom of the screen. Numbers are shown for the threat bar to give a more accurate idea of its progress, but during gameplay the colour changing of the threat bar provides a quicker and easier way for the player to keep track of how bad things are getting at a glance. To allow the player to make informed decisions, we used the concept of an IT worker receiving notifications on their phone and having to read and assess where they were needed most.

Initially, the notification of an issue shows up on the top center of the screen with the name and issue description through an employee's perspective. This keeps the message organic and grounds it to how it would appear in real life. Rather than saying 'there is a malicious link, should I click it?' the message would say 'I've gotten a request from the marketing department to drop our sales report ahead of time in this dropbox link as they need it fast, should I do it?' Issues are not always straightforward in intent, and even a well-intentioned message may still mask a malicious issue.

Once the notification appears in the center screen, it then moves to the player's phone in the bottom right corner. The messages appear as chat messages that show the time, employee name, and a brief portion of the issue. Clicking on the issue allows the player to expand it for the full

message, and click again to shrink it. Issues are dynamically sorted by time remaining, with the least time shown at the top. The phone is the primary way to keep the player informed of multiple ongoing issues in an organized manner.

The HUD's main controller script was `OfficeUI.gd`, attached to a `CanvasLayer` on the main gameplay screen. The `CanvasLayer` ensures all HUD elements float in fixed positions on screen. Beyond the simple score, timer, and threat bar elements, the HUD contained three more complex components: the `IssuePopup` tablet, the center `Toaster Banner`, and the phone notification list.

The `Toaster Banner` appears center screen when a new issue spawns. It is its own independent scene called via `showtoast()` from `OfficeUI.gd`, passing the employee name and current issue message for display. At the same time, the issue is added to the phone's `NotificationList`. A `VBoxContainer` organizes the `NotificationItem` scenes, which display the name, remaining time, and a 45-character preview of the message. The dynamic time-based sorting is handled by a dedicated script on the `NotificationList` node. `NotificationItems` can be expanded and collapsed by clicking to show the full message without cluttering the phone when multiple issues are active simultaneously.

Popup UI: iPad-Style Tablet

The incident popup was designed to resemble an Apple iPad in dark mode to reinforce the technology setting of the game. When the player presses E near an NPC with an active issue, the tablet slides up from the bottom of the screen using a cubic ease-out tween animation. The scene contains a dark rounded bezel with a camera bar pill at the top, a status bar showing the app name and time, a blue accent line below the header, a scrollable body with an `INCIDENT` section showing the employee's message and a `SELECT RESPONSE` section with three answer buttons, a footer with a `Close Ticket` button, and a home indicator pill at the bottom.

Button styling is applied at runtime in `issue_popup.gd._style_buttons()` using `StyleBoxFlat` objects. Buttons remain neutral in appearance regardless of whether the answer was correct or incorrect; only the feedback text changes, avoiding colour cues that might oversimplify the learning moment. Wrong answers do not highlight the correct answer in green -the player must read the explanation to understand what the correct response was.

Audio System

An `AudioManager` singleton autoload handles all music and sound effect playback. Music plays in a continuous loop using a finished-signal reconnect approach, since setting `AudioStreamWAV.loop_mode` at runtime proved unreliable across Godot versions. The finished signal reconnects the `play()` call, effectively creating a manual loop that works regardless of import settings.

Five sound effects were generated procedurally using Python's wave and struct modules: a rising typewriter chime for correct answers, a descending two-tone buzz for wrong answers, a double notification blip for popup opens, a four-cycle high-low alarm for escalation events, and a sharp clock tick for the timer warning at 20 seconds remaining. Two music tracks were generated: an upbeat 108 BPM lo-fi office jazz piece for general gameplay, and a 160 BPM intense minor-key track using square wave bass, dissonant tritone stabs, and double kick patterns for the lockdown event.

Cutscene System

Two cutscenes were implemented using a shared pattern. The intro cutscene plays before the shift begins, delivering lines including the typewriter phrase 'Grab your coffee, kid. Good luck.' Each character is appended to the label text on a timer interval of 0.04 seconds, with a 1.2-second pause between lines. A skip button allows immediate transition to the game. The game over transition cutscene selects different message sets based on performance: a perfect run, a failed run where more incidents failed than were handled, a company-compromised run where the danger bar hit 100, or an average run. The cutscene uses Tween-based delays rather than await coroutines to avoid signal chain errors that caused crashes in earlier iterations.

Reflection Report

The Reflection Report was built as a standalone Godot scene (ReflectionReport.tscn) that loads after the game over screen. It reads from GameManager.wrong_answers, an array that gets populated in two ways: when a player selects an incorrect answer in the tablet popup, issue_popup.gd calls GameManager.record_wrong_answer() passing the issue name, correct answer, explanation, and the original question description. The second path is through the escalation system -when an NPC's patience runs out and escalate() fires in employee.gd, that also calls record_wrong_answer() so jobs the player ignored or did not reach in time show up in the report too, not just ones they actively answered wrong.

The report itself is an iPad-styled scene that mirrors the tablet popup aesthetic. Each wrong answer renders as a card built in code using VBoxContainer nodes created at runtime in ReflectionReport.gd. Each card shows the issue name, the original question the player saw, the correct answer highlighted, and an explanation of why that response was right. The cards are generated in a loop from the wrong_answers array, so the report is always specific to that session with no repeated entries -record_wrong_answer() checks for duplicates before appending.

The scene slides in from the right side of the screen using the same cubic ease tween approach as the tablet popup for visual consistency. A close button returns the player to the game over screen. On Play Again, GameManager.wrong_answers is cleared along with the rest of the game state, so each new session starts with a fresh report.

What Worked

The signal-based architecture in GDScript kept the codebase modular and allowed us to work across separate branches without frequent merge conflicts. The resource file approach for cybersecurity scenarios made it possible to add 24 new issues in a single branch without modifying any gameplay script. The singleton GameManager pattern provided clean shared state management. The tablet popup animation and the lockdown event were consistently identified as polished features in playtesting. The procedurally generated audio assets matched the game's aesthetic without requiring external audio licensing.

What Did Not Work

Several significant technical challenges were encountered. NPC invisibility persisted across multiple development sessions and was ultimately traced to sprite atlas `SPRITE_BASE_X` values exceeding the pixel width of the characters.png sheet, causing certain NPC types to sample transparent regions. The fix required remapping all base X values to fit the actual sheet dimensions.

The Area2D interaction detection system for determining which NPC the player was standing near proved unreliable when employee collision layers were modified. After several iterations using collision layer separation, the final solution reverted to the original layer 1 configuration that our original codebase used, restoring reliable detection.

The lockdown event silently cancelled itself for multiple sessions because the LockdownEvent node had been added to the working build's office.tscn but not to our branch's office.tscn. The node existed in the scene definition but never instantiated, causing `find_child()` to return null and the handler to return early after resetting lockdown_active. This was identified by tracing the null reference error through the debugger call stack.

WAV file looping required a workaround because `AudioStreamWAV.loop_mode` set at runtime through a script is not reliably applied in Godot 4.6/4.7. The finished signal reconnect approach resolved this.

Our initial design lacked proper response feedback to negative player actions, leaving it unclear in the moment whether a decision was right or wrong. The threat bar was implemented to give better real-time feedback, with colour coding making the severity immediately visible. The Reflection Report was added to provide a more traditional post-action review. The reflection report initially did not count timed-out answers as wrong, which was later corrected.

NPC movement caused issues with NPCs getting stuck or behaving erratically, so movement was brought to a minimum. The map design also did not account for how often NPCs might unintentionally trap players or each other.

The scoreboard did not accumulate or add points correctly and would not connect. The code was rewritten so that there was a clear value attached to each scenario, the game already had the correct answers stored, and the scoreboard correctly added points received while subtracting for mistakes and then outputting the result.

Results and Outcomes

Playtesting Methodology

A playtesting study was conducted with 9 participants, 3 recruited by each team member from their personal networks. All testers were between 20 and 35 years old. 6 were students and 3 were working full-time. In terms of gaming background, 4 primarily play on PC, 3 on console, and 2 on mobile. Each session consisted of one full gameplay shift without coaching or hints. Participants were observed playing and asked to complete a short survey after their session.

Engagement and Difficulty

6 of 9 testers said the game was engaging, and the same 6 said they would play it again. Difficulty ratings were spread across the scale: 2 testers rated it a 1, 2 rated it a 2, 3 rated it a 3, and 2 rated it a 5, suggesting the game sits at a comfortable range for most players without feeling too easy or too punishing. The spread also reflects the varying gaming backgrounds in the group -more experienced players tended to rate the difficulty lower.

Learning Outcomes

8 of 9 testers said they felt they learned something new after their session, which was the strongest result from the survey. More notably, all 9 testers said they found the game more engaging as a learning method than a traditional classroom setting. This is the clearest validation of the project's core premise -that active, consequence-driven gameplay is a more effective format for cybersecurity awareness than passive instruction.

Lockdown Event Outcomes

5 of 9 testers encountered the lockdown event during their session. The lockdown was consistently mentioned in open responses as the moment that caught their attention the most, with several testers describing it as the most stressful and memorable part of the game. The remaining 4 testers who did not encounter it noted the standard incident responses and the danger bar rising as the most noticeable moments.

Session Outcomes

4 testers ended their shift when the timer ran out, while 5 ended due to the threat bar filling up completely. The fact that most testers hit the threat bar rather than running down the clock suggests the danger accumulation system is working as intended -wrong answers and timed-out jobs are creating enough consequence to end sessions before the shift naturally expires.

Reflection Report Engagement

6 of 9 testers opened and read the Reflection Report at the end of their session. For a feature that players are not required to interact with, a two-thirds engagement rate suggests the post-shift

review is compelling enough to explore without prompting. The 3 testers who did not read it either ended on a game over and moved quickly past the screen, or missed it entirely.

Observations

The most significant finding from the survey was the unanimous agreement that game-based learning felt more engaging than a classroom setting. 9 out of 9 testers responded yes to this question, which directly supports the motivation behind the project. Even testers who did not find the game particularly engaging overall still preferred it as a learning format over traditional instruction.

The lockdown event stood out as the most discussed feature in open responses. Testers who encountered it described it as the moment the game felt most real and most stressful. This aligns with the design intent -the 45-second timer and the need to physically clear every NPC was meant to simulate the pressure of real incident response, and the feedback confirms it landed that way.

The difficulty spread was worth noting. No single rating dominated, which suggests the game scales reasonably across different experience levels. More experienced gamers found it straightforward while less experienced players found it more challenging, but neither group felt it was unfair.

The one tester who said they did not learn anything new was from a technical background, which is consistent with the expectation that the lower and mid-tier scenarios would feel familiar to someone already working in or studying IT. The critical threat scenarios, however, generated surprise even among technically aware testers.

Conclusions and Future Work

Conclusions

Reply All demonstrates that a lightweight, accessible game can deliver meaningful cybersecurity awareness content to non-technical organizational employees without requiring prior security knowledge or gaming experience. The combination of realistic workplace scenarios written from the employee's perspective, consequence-driven mechanics where wrong answers raise a danger bar and cause escalations, time pressure during critical events, and post-session reflection created a learning loop that our playtesting suggests is more engaging than passive training formats for security decision-making.

The most significant result from our 9-tester survey was that all 9 participants said they found game-based learning more engaging than a traditional classroom setting. 8 of 9 said they learned something new from their session, and 6 of 9 said they would play the game again. These numbers hold up across both students and working professionals in the group, suggesting the format is accessible regardless of background.

The lockdown event was consistently highlighted in open responses as the most memorable moment in the game. 5 of the 9 testers encountered it, and those who did described it as the point where the game felt most real. This supports the idea that high-pressure, consequence-driven moments reinforce memory in ways that standard question-and-answer formats do not. The fact that 5 of 9 sessions ended with the threat bar filling up rather than the timer running out also confirms the danger system is working as intended - wrong decisions accumulate into real consequences.

The Reflection Report was read by 6 of 9 testers without being prompted, which for an optional post-game screen is a strong engagement rate. Combined with the 8 of 9 learning result, it suggests the post-shift review adds educational value beyond the gameplay itself.

From a technical standpoint, Godot 4.6/4.7's signal architecture, resource-based content system, and GDScript proved well-suited to rapid collaborative development. The main friction came from asset pipeline management, particularly sprite sheet atlas coordinates and WAV looping behavior, and from branch integration issues when scene file changes in one branch were not carried over to the merged build. Both were resolved through systematic debugging and cleaner merge procedures.

Future Work

Several extensions were identified during development that were deferred due to timeline constraints:

- Formal pre- and post-session knowledge assessment to produce quantitative evidence of learning gain from a single gameplay session, enabling comparison against a control group completing equivalent time in standard awareness training.
- A streak-based reward and debuff system that provides the player with bonus points for consecutive correct answers and increased danger bar drain rate after consecutive failures, creating a more dynamic difficulty curve.
- An expanded NPC roster with unique sprite designs for each character type and additional role-specific patience and movement configurations.
- A web export to HTML5 via Godot's export pipeline to allow browser-based access without local installation, enabling broader organizational pilots.
- A scenario authoring tool allowing non-developer administrators to create new issue .tres files through a web interface, enabling organizations to add domain-specific or role-specific incident scenarios without modifying the main game code.
- A multiplayer coordination mode in which two players manage different office areas simultaneously and must communicate to prioritize which incidents to address first.
- Further branching jobs that add more scenarios based on previous failed jobs or certain wrong answer choices.
- Updated sprites that change based on score milestones for the player, like a promotion system.
- More SFX for each button press on menus and phone notifications.
- Further balancing changes and scenario updates based on survey feedback from a larger sample size.

Bibliography

- [1] SentinelOne, "Data Breach Statistics," SentinelOne Cybersecurity 101, 2024. [Online]. Available:
<https://www.sentinelone.com/cybersecurity-101/cybersecurity/data-breach-statistics/>
- [2] K. Roer and G. Petric, "Security Culture Report 2022," CLTRe/KnowBe4, 2022.
- [3] M. Hendrix, A. Al-Sherbaz, and V. Bloom, "Game Based Cyber Security Training: are Serious Games suitable for cyber security training?" *Int. J. Serious Games*, vol. 3, no. 1, 2016.
- [4] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, "From game design elements to gamefulness: defining gamification," in *Proc. 15th Int. Academic MindTrek Conf.*, Tampere, Finland, 2011, pp. 9-15.
- [5] R. E. Mayer, "Computer games in education," *Annual Review of Psychology*, vol. 70, pp. 531-549, 2019.
- [6] R. Heartfield and G. Loukas, "A Taxonomy of Attacks and a Survey of Defence Mechanisms for Semantic Social Engineering Attacks," *ACM Computing Surveys*, vol. 48, no. 3, Art. 37, 2015.
- [7] M. Workman, "Wisecrackers: A theory-grounded investigation of phishing and pretext social engineering threats to information security," *J. American Society for Information Science and Technology*, vol. 59, no. 4, pp. 662-674, 2008.
- [8] Proofpoint, "State of the Phish 2023," Proofpoint Inc., 2023. [Online]. Available:
<https://www.proofpoint.com/us/resources/threat-reports/state-of-phish>
- [9] C. Irvine, M. Thompson, and K. Allen, "CyberCIEGE: gaming for information assurance," *IEEE Security and Privacy*, vol. 3, no. 3, pp. 61-64, May-Jun. 2005.
- [10] P. Moreno-Ger, D. Burgos, I. Martinez-Ortiz, J. T. Sierra, and B. Fernandez-Manjon, "Educational game design for online education," *Computers in Human Behavior*, vol. 24, no. 6, pp. 2530-2540, 2008.

Appendices

Appendix A: Project Links

GitHub Repository: <https://github.com/Sonarc-sys/Reply-All>

Godot Version: 4.6.1 stable / 4.7.1 stable

Appendix B: Full Issue Catalog

All 36 cybersecurity scenarios implemented in Reply All, stored as .tres resource files in resources/Issue Resources/:

- Low (1-2): ForgotPassword, WeakPassword, PublicWifi, UnlockedScreen, SharedPassword, TailgatingAccess
- Mid (3): PhishingEmail, SocialEngineering, PrinterData, SuspiciousUSB, SoftwareUpdate, MaliciousAttachment, RemoteDesktopExposed, CloudMisconfiguration
- High (4): DataLeak, UnauthorisedAccess, RansomwareAlert, ZeroDayVuln, InsiderDataTheft, BECAAttack
- Critical (5): ActiveIntrusion, CredentialDump, RansomwareDeployed, SupplyChainAttack, MFABypass, C2Detection, DataExfiltrationLive
- Team's original issues also included in game: CEOSpoofing, Cleartext, Cryptojacking, EmailWorm, FoundUSB, RansomwarePropagation, ShadowIT, Spear Phishing, Tailgating

Appendix C: Playtesting Data Summary

Tester Demographics

- Total testers: 9
- Age range: 20 to 35 years old
- Students: 6 | Working full-time: 3
- Gaming platform: PC (4), Console (3), Mobile (2)

Engagement Survey Results

- Found the game engaging: 6/9 (67%)
- Would play again: 6/9 (67%)
- Preferred game-based learning over classroom: 9/9 (100%)
- Felt they learned something new: 8/9 (89%)

Difficulty Ratings (1 = easiest, 5 = hardest)

- Rated 1: 2 testers | Rated 2: 2 testers | Rated 3: 3 testers | Rated 4: 0 testers | Rated 5: 2 testers

Session Outcomes

- Shift ended by timer running out: 4/9
- Shift ended by threat bar filling up: 5/9

Lockdown Event

- Testers who encountered the lockdown event: 5/9
- Testers who did not encounter it: 4/9

Reflection Report

- Testers who read the Reflection Report: 6/9 (67%)
- Testers who felt they learned something new: 8/9 (89%)

Appendix D: Individual Contribution Summary

Troy Patrick: SFX generation and AudioManager autoload, iPad tablet popup UI and slide-in animation, intro and game over cutscene typewriter system, Options menu with volume and shift length controls, critical breach lockdown event and LockdownEvent scene, 24 new cybersecurity issue .tres files, Reflection Report scene, NPC collision debugging and sprite atlas fix, employee.gd patience tuning.

Vladimir Gappasov: Score and points system, threat level danger bar with colour-coded ProgressBar, shift timer label and update system, scoreboard scene integration, issue content contributions.

Ayinkaran Ravindran: Character sprite sheet improvements, map design and Y-sorting for visual depth, CanvasItem layer HUD elements and OfficeUI.gd coding, NPC urgency timing per role, escalation consequence system in employee.gd, walking animation frame configuration, notification system for toaster banner and phone.